

Visualizing image and Histogram with openCV

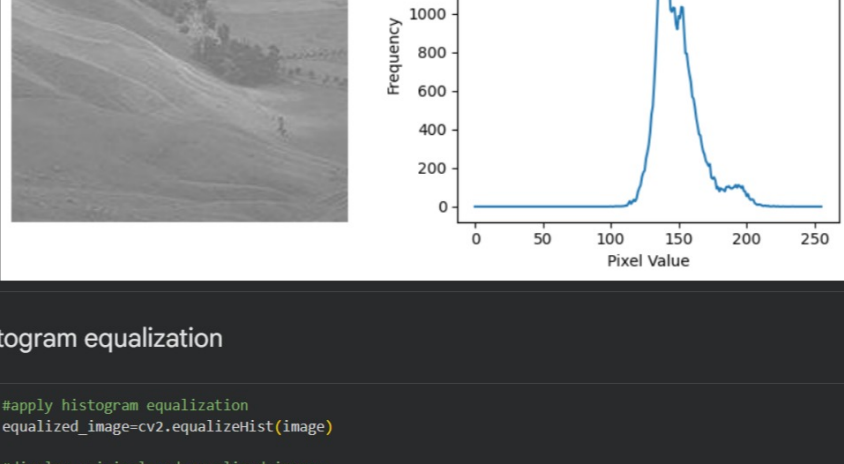
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab.patches import cv2_imshow

#load the image
image=cv2.imread("image-with-bad-contrast_0320.jpg",cv2.IMREAD_GRAYSCALE)
#calculate the histogram
histogram=cv2.calcHist([image],[0],None,[256],[0,256])

#display the original image
plt.figure(figsize=(8,4))
#plt.subplot( rows,columns,index)
#plt.subplot(1,2,1) -> [ACTIVE |
#plt.subplot(1,2,1) -> [ | ACTIVE ]
plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(image,cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

#display the histogram
plt.subplot(1,2,2)
plt.plot(histogram)
plt.title("Histogram")
plt.xlabel("Pixel Value")
plt.ylabel("Frequency")

#show the plots
plt.tight_layout()
plt.show()
```



Histogram equalization

```
#apply Histogram equalization
equalized_image=cv2.equalizeHist(image)

#display original and equalized images
print ("Original Image")
cv2_imshow(image)
print ("Equalized Image")
cv2_imshow(equalized_image)

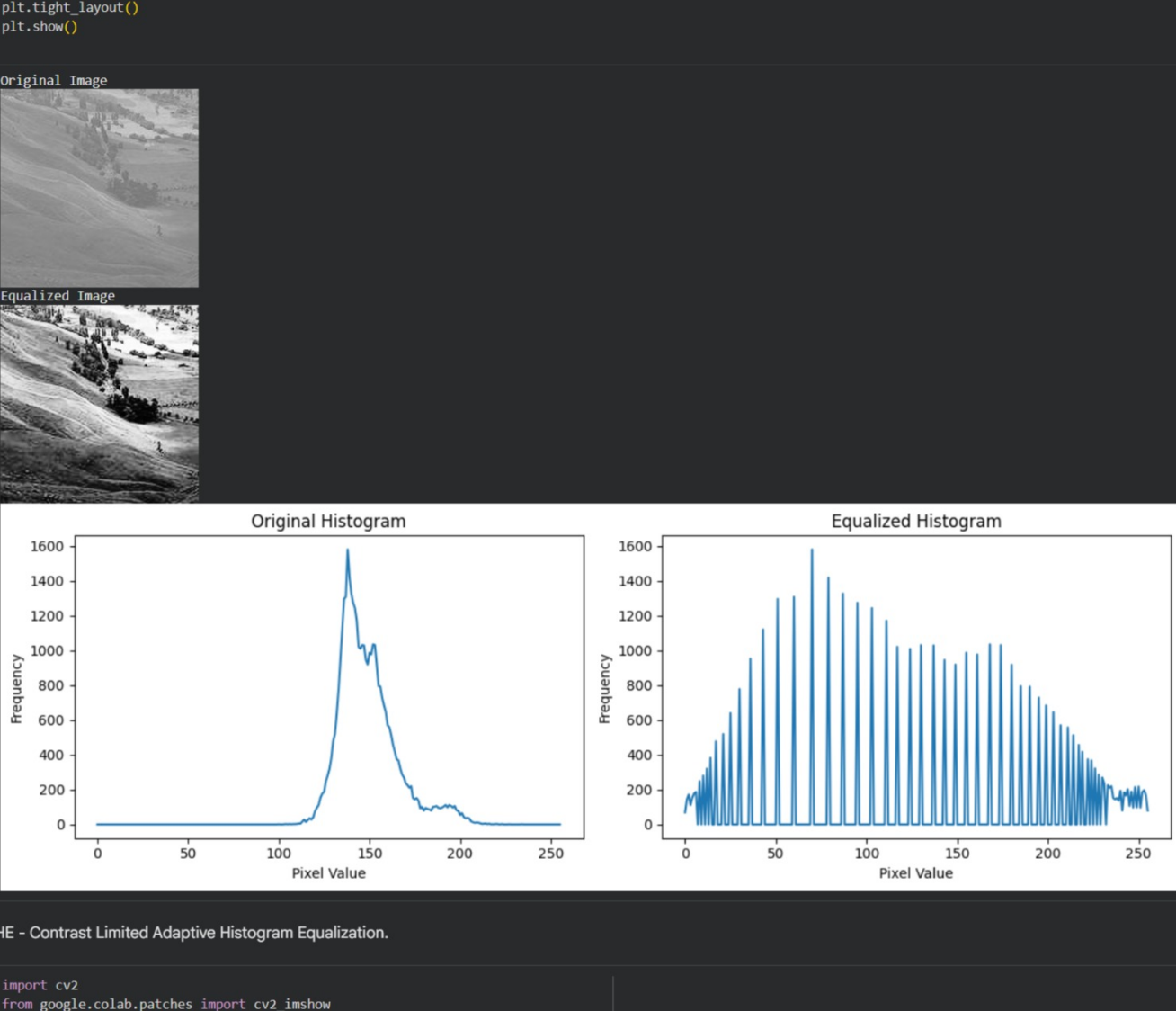
#calculate histograms
hist_original=cv2.calcHist([image],[0],None,[256],[0,256])
hist_equalized=cv2.calcHist([equalized_image],[0],None,[256],[0,256])

#plot histograms side by side
plt.figure(figsize=(12,4))

#Original Histogram
plt.subplot(1,2,1)
plt.plot(hist_original)
plt.title("Original Histogram")
plt.xlabel("Pixel Value")
plt.ylabel("Frequency")

#equalized histogram
plt.subplot(1,2,2)
plt.plot(hist_equalized)
plt.title("Equalized Histogram")
plt.xlabel("Pixel Value")
plt.ylabel("Frequency")

#show the plots
plt.tight_layout()
plt.show()
```



CLAHE - Contrast Limited Adaptive Histogram Equalization.

```
import cv2
from google.colab.patches import cv2_imshow

#load the image in grayscale mode
#Q0tql.png - file name
#cv2.imread_grayscale - forces the image to be loaded as a single-channel
img = cv2.imread('Q0tql.png',cv2.IMREAD_GRAYSCALE)

assert img is not None, "file could not be read, check with os.path.exists()"

#display the original image
cv2_imshow(img)

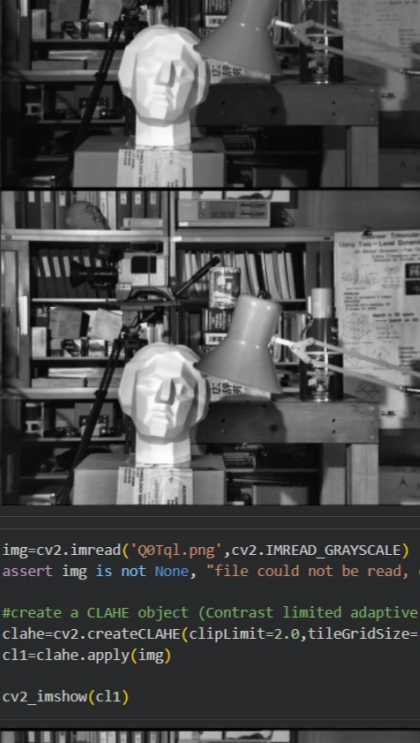
#create a CLAHE object (Contrast limited adaptive histogram equalization)
#clipLimit=2.0 -> limits how much contrast is increased (prevents noise amplification)
#tileGridSize=(8,8) -> divides the image into 8*8 small regions (tiles)

clahe=cv2.createCLAHE(clipLimit=2.0,tileGridSize=(8,8))

#apply the CLAHE to the grayscale image
#this enhances local contrast in each tile while controlling over-enhancement
c1i=clahe.apply(img)

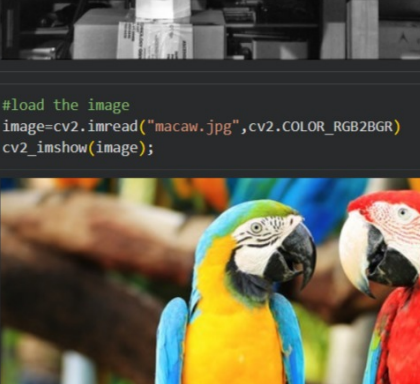
#display the enhanced image after applying CLAHE
cv2_imshow(c1i)

#show the processed image to disk
# cv2_imshow()
```



```
img=cv2.imread('Q0tql.png',cv2.IMREAD_GRAYSCALE)
assert img is not None, "file could not be read, check with os.path.exists()"

#create a CLAHE object (Contrast limited adaptive histogram equalization)
clahe=cv2.createCLAHE(clipLimit=2.0,tileGridSize=(8,8))
c1i=clahe.apply(img)
```



```
#load the image
image=cv2.imread("macaw.jpg",cv2.COLOR_RGB2BGR)
cv2_imshow(image);
```

```
B=image[:, :,0] #blue layer
G=image[:, :,1] #green layer
R=image[:, :,2] #red layer

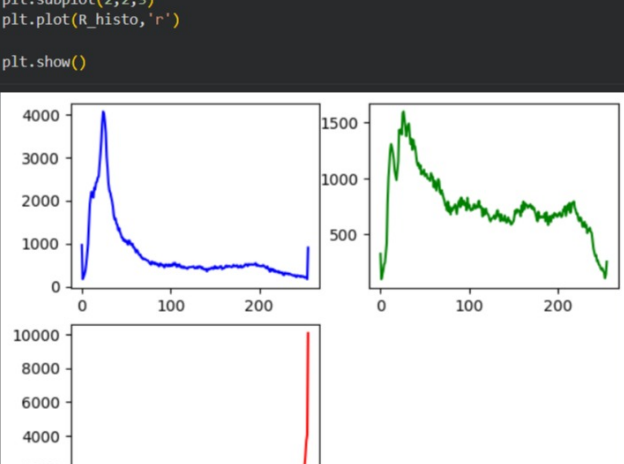
B_histo=cv2.calcHist([B],[0],None,[256],[0,256])
G_histo=cv2.calcHist([G],[0],None,[256],[0,256])
R_histo=cv2.calcHist([R],[0],None,[256],[0,256])

plt.subplot(2,2,1)
plt.plot(B_histo, 'b')

plt.subplot(2,2,2)
plt.plot(G_histo, 'g')

plt.subplot(2,2,3)
plt.plot(R_histo, 'r')

plt.show()
```



Low Contrast Image

```
#load the image
image=cv2.imread("macaw_lc.jpg",cv2.COLOR_RGB2BGR)
cv2_imshow(image);
```



```
B=image[:, :,0] #blue layer
G=image[:, :,1] #green layer
R=image[:, :,2] #red layer
```

```
#equalize
B_eq=cv2.equalizeHist(B)
G_eq=cv2.equalizeHist(G)
R_eq=cv2.equalizeHist(R)
```

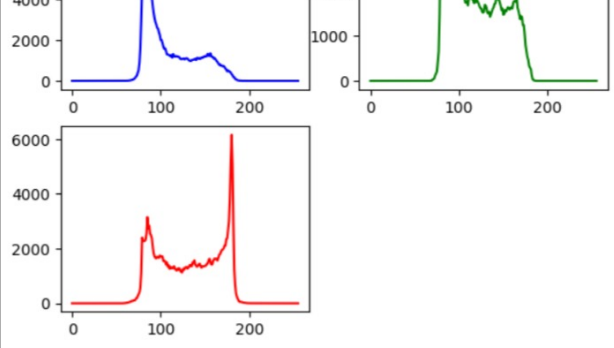
```
#histogram
B_histo=cv2.calcHist([B],[0],None,[256],[0,256])
G_histo=cv2.calcHist([G],[0],None,[256],[0,256])
R_histo=cv2.calcHist([R],[0],None,[256],[0,256])
```

```
#show histograms
plt.subplot(2,2,1)
plt.plot(B_histo, 'b')

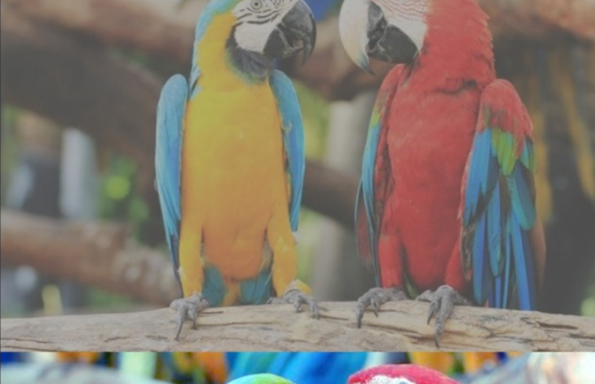
plt.subplot(2,2,2)
plt.plot(G_histo, 'g')

plt.subplot(2,2,3)
plt.plot(R_histo, 'r')

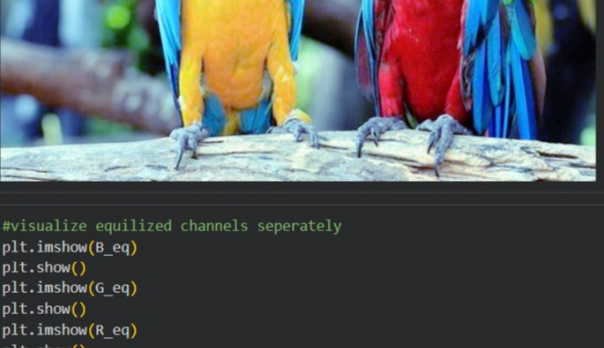
plt.show()
```



```
#merge the channels and create new image
image_eq=cv2.merge([B_eq,G_eq,R_eq])
cv2_imshow(image_eq)
```



```
#visualize equalized channels separately
plt.imshow(B_eq)
plt.show()
plt.imshow(G_eq)
plt.show()
plt.imshow(R_eq)
plt.show()
```



```
#equalized histograms
B_histo_eq=cv2.calcHist([B_eq],[0],None,[256],[0,256])
G_histo_eq=cv2.calcHist([G_eq],[0],None,[256],[0,256])
R_histo_eq=cv2.calcHist([R_eq],[0],None,[256],[0,256])
```

```
#show histograms
plt.subplot(2,2,1)
plt.plot(B_histo_eq, 'b')

plt.subplot(2,2,2)
plt.plot(G_histo_eq, 'g')

plt.subplot(2,2,3)
plt.plot(R_histo_eq, 'r')
```

```
[cmatplotlib.figure.Figure at 0x7dbdc6decf0]
```

